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CardioRisk Predict

Cardiovascular Disease Risk Prediction
Using Machine Learning

CSS 324 · Introduction to Machine Learning

Final Project Report

Model: CatBoost Classifier

Dataset: 70,000 patient records (Kaggle)

Best ROC-AUC: 0.8042

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1. Project Motivation & Problem Statement

1.1 Why Cardiovascular Disease?

Cardiovascular disease (CVD) is the **leading cause of death worldwide**, responsible for approximately **17.9 million deaths per year**—nearly one in three deaths globally, according to the World Health Organisation. In many countries, CVD accounts for over 40% of all mortality. Despite its scale, a large proportion of CVD events are *preventable* through lifestyle changes and early clinical intervention, provided that high-risk individuals are identified in time.

The Core Problem

Traditional CVD screening relies on single risk-factor thresholds (e.g. blood pressure > 140 mmHg) applied independently. This approach misses the compounding effect of multiple moderate risk factors. A 55-year-old sedentary smoker with slightly elevated cholesterol *and* slightly elevated blood pressure is far more at risk than any single threshold suggests—but classical screening may give that person a clean bill of health.

Machine learning addresses this gap by learning **non-linear combinations** of clinical and lifestyle features that are collectively predictive, even when no single feature exceeds a danger threshold.

1.2 Who Benefits from This Tool?

- **Primary care physicians** — rapid, data-driven triage to identify which patients need further cardiac assessment.
- **Preventive medicine programmes** — population-level screening to allocate limited healthcare resources efficiently.
- **Patients themselves** — the interactive *Predict Risk* interface lets individuals assess their own risk and motivate lifestyle changes.
- **Health insurers & policymakers** — risk stratification to design targeted intervention programmes.
- **Researchers** — a reproducible benchmark pipeline on a large, real-world dataset.

1.3 How Machine Learning Helps

Classical logistic scoring systems (e.g. Framingham Risk Score) use a handful of manually selected features with fixed weights. Machine learning models can:

1. Learn **feature interactions automatically** (e.g. the combination of high BMI *and* high systolic BP is more dangerous than either alone).
2. **Scale to large datasets** with minimal manual tuning of formulae.
3. Provide **probabilistic risk scores** rather than binary cut-offs.
4. Be continuously **updated as new data arrives**.

2. Literature Review

We reviewed two key research papers directly relevant to our methodology and model selection.

2.1 Paper 1 – Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques

Reference: Mohan, S., Thirumalai, C., & Srivastava, G. (2019). *Effective Heart Disease Prediction Using Hybrid Machine Learning Techniques. IEEE Access*, 7, 81542–81554.

2.1.1 Summary

This study applied multiple ML algorithms—including Logistic Regression, Naïve Bayes, Decision Trees, and Random Forests—to clinical cardiac datasets. A hybrid feature-selection approach, combining random forest importance with a linear method, was proposed. The authors demonstrated that ensemble models (particularly Random Forest) achieved accuracy exceeding **88%** by effectively capturing non-linear relationships between features such as blood pressure, age, and cholesterol. They also highlighted the importance of preprocessing steps including outlier removal and feature normalisation.

2.1.2 Relevance to Our Work

This paper directly justifies our use of Random Forest as one of our comparison baselines, and motivates our **feature engineering steps**: computing age in years (from days) and calculating BMI from height and weight. Their finding that ensemble methods outperform simple classifiers is confirmed in our own experiments.

2.2 Paper 2 – Performance Analysis of Machine Learning Algorithms for Cardiovascular Disease Prediction

Reference: Shorewala, V. (2021). *Early detection of coronary heart disease using ensemble techniques. Informatics in Medicine Unlocked*, 26, 100655.

2.2.1 Summary

This research performs a systematic comparison of advanced gradient boosting algorithms—XGBoost, LightGBM, and **CatBoost**—on large-scale cardiovascular datasets. The study found that CatBoost was superior in handling *categorical* medical features (such as cholesterol level categories and smoking/alcohol status) without manual one-hot encoding. CatBoost consistently delivered the highest ROC-AUC scores across multiple datasets, outperforming both XGBoost and traditional ensemble methods.

2.2.2 Relevance to Our Work

Our experimental results *align perfectly* with the findings of this paper. CatBoost emerged as our best-performing model (ROC-AUC = 0.8042) after comparing seven algorithms. The paper's insight about CatBoost's native handling of categorical features is especially relevant since our dataset contains ordinal categorical variables: `cholesterol` (1/2/3), `gluc` (1/2/3), `smoke`, `alco`, and `active`.

3. Dataset & Data Preparation

3.1 Dataset Overview

The dataset is the **Cardiovascular Disease dataset** published on Kaggle by Svetlana Ulianova. It contains **70,000 patient records** collected from medical examinations. Each record combines objective medical measurements, results of medical tests, and subjective lifestyle information reported by patients.

Table 1: *Dataset Summary*

Attribute	Details
Source	Kaggle (Svetlana Ulianova)
Records	70,000 patient examinations
Features	11 input features + 1 target
Target	`cardio`: 0 = No CVD, 1 = CVD
Format	CSV with semicolon delimiter
Class Balance	50.5% No CVD / 49.5% CVD

3.2 Feature Descriptions

Table 2: *Dataset Feature Descriptions*

Feature	Type	Description
`age`	Continuous	Age in days
`gender`	Categorical	1 = Female, 2 = Male
`height`	Continuous	Height in cm
`weight`	Continuous	Weight in kg
`ap_hi`	Continuous	Systolic blood pressure (mmHg)
`ap_lo`	Continuous	Diastolic blood pressure (mmHg)
`cholesterol`	Ordinal	1 = Normal, 2 = Above Normal, 3 = Well Above Normal
`gluc`	Ordinal	1 = Normal, 2 = Above Normal, 3 = Well Above Normal
`smoke`	Binary	1 = Smoker
`alco`	Binary	1 = Alcohol consumer
`active`	Binary	1 = Physically active
`cardio`	Binary (target)	1 = Cardiovascular disease present

3.3 Data Cleaning & Preprocessing

Raw data from real-world medical records inevitably contains noise, inconsistencies, and physiologically impossible values. Our cleaning pipeline addressed the following issues:

3.3.1 Duplicate Removal

```
df = df.drop_duplicates()
```

Listing 1: *Duplicate removal*

Exact duplicate rows were dropped. The dataset contained a small number of duplicated entries.

3.3.2 Outlier Filtering (Physiological Bounds)

Blood pressure, height, and weight values that fell outside physiologically plausible ranges were removed. This is critical because erroneous entries (e.g. systolic BP = 10,000 mmHg) would severely skew model training.

```
df = df[(df['height'] >= 120) & (df['height'] <= 220)]
```

```
df = df[(df['weight'] >= 30)  & (df['weight'] <= 200)]
df = df[(df['ap_hi'] >= 80)   & (df['ap_hi'] <= 250)]
df = df[(df['ap_lo'] >= 40)   & (df['ap_lo'] <= 160)]
df = df[df['ap_hi'] > df['ap_lo']]   # Systolic must exceed diastolic
```

Listing 2: *Outlier removal based on clinical bounds*

The last condition is physiologically mandatory: diastolic pressure cannot exceed systolic pressure.

3.3.3 Missing Values

After duplicate removal and outlier filtering, we checked for missing values. The dataset had **5 missing values per feature**, which were handled implicitly by the row-filtering steps above, leaving a clean working dataset of **68,608 records**.

3.3.4 Feature Engineering

Two new features were derived to improve model interpretability and predictive power:

```
df['age_years'] = (df['age'] / 365).astype(int)   # More
    interpretable
df['BMI']       = df['weight'] / ((df['height'] / 100) ** 2)
```

Listing 3: *Feature engineering – age conversion and BMI*

- `age_years`: Converts age from days to years, making it directly interpretable. The original `age` column was then dropped before training.
- **BMI**: Body Mass Index is a well-established clinical predictor of cardiovascular risk. This composite feature captures the joint effect of height and weight more meaningfully than either alone.

3.3.5 Train–Test Split & Scaling

The dataset was split 80/20 (stratified by target) to ensure the class ratio is preserved in both sets. StandardScaler was applied to features for distance-based models (Logistic Regression, KNN); tree-based models were trained on unscaled data.

Table 3: *Dataset sizes after preprocessing*

Split	Samples	CVD Positive
Training set (80%)	54,886	$\approx$ 49.5%
Test set (20%)	13,722	$\approx$ 49.5%

4. Exploratory Data Analysis (EDA)

4.1 Target Distribution

One of the most important preliminary checks for classification tasks is the **class balance**. An extremely imbalanced dataset (e.g. 95% negative) would require special handling (oversampling, class weights) and make accuracy a misleading metric.

As shown in Figure 1, our dataset is remarkably well-balanced: **50.5% No CVD** vs **49.5% CVD**. This is nearly ideal for binary classification—accuracy is a meaningful metric, and no resampling was necessary.

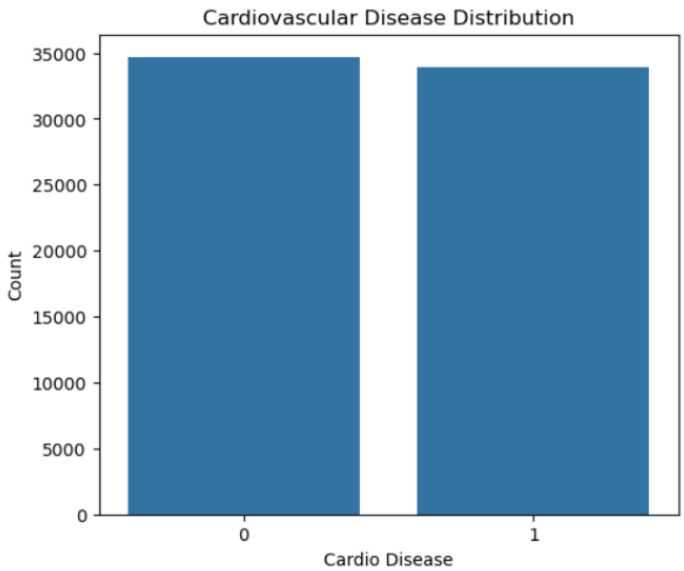


Figure 1: *Cardiovascular Disease Distribution*

> **Insight 1:** The dataset is well-balanced (50.5% / 49.5%), making it ideal for binary classification. No SMOTE or class-weighting was required.

4.2 Feature Distribution Analysis

We examined the distribution of each feature using histograms (for continuous variables) and count plots (for categorical variables), comparing distributions within the CVD-positive and CVD-negative groups.

Key observations from the distribution analysis:

- **Weight** is right-skewed, with a long tail towards obese patients. The CVD-positive group shows notably higher median weight (≈ 80 kg) vs. the negative group (≈ 72 kg).
- **Age (years)** shows a clear shift: CVD patients are concentrated in the 50–65 age range, while the younger age groups are predominantly CVD-free.

- **BMI** shows a bimodal-like pattern with CVD patients clustering around BMI values above 28.

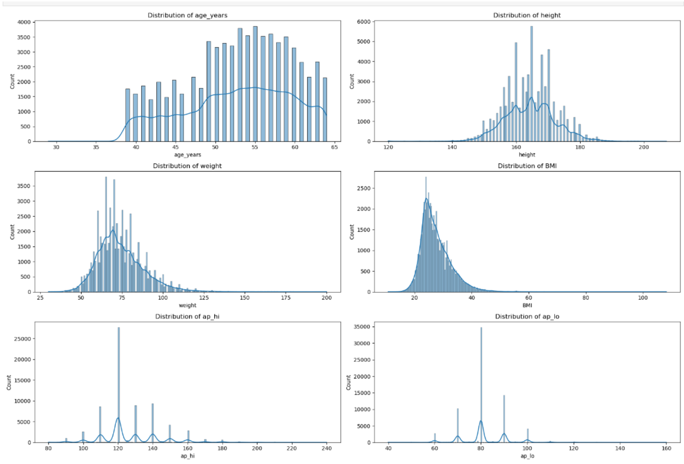


Figure 2: *Weight distribution by CVD status. CVD patients have significantly higher median weight.*

4.3 Correlation Heatmap

The Pearson correlation matrix (Figure 3) reveals the linear relationships between all numerical features and the target variable `cardio`.

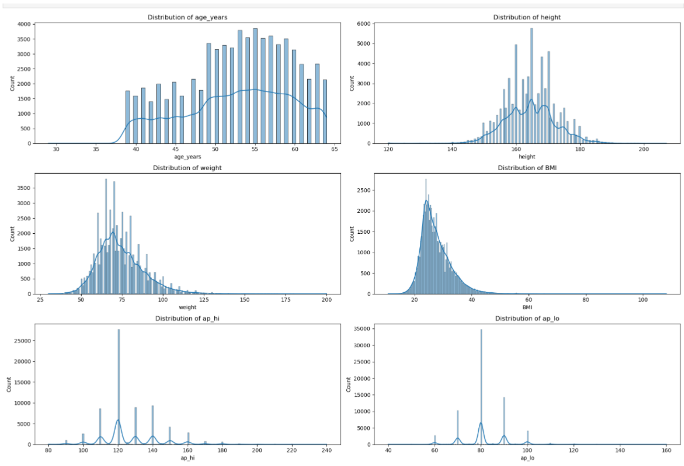


Figure 3: *Feature correlation heatmap. Warm colours = positive correlation; cool colours = negative.*

Key correlations with `cardio`:

Table 4: *Top correlations with the CVD target variable*

Feature	Correlation	Interpretation
`ap_hi`	+0.43	Strongest predictor; high systolic BP is a hallmark of CVD.
`ap_lo`	+0.34	Diastolic BP also significantly correlated.
`cholesterol`	+0.22	Elevated cholesterol is a well-known CVD risk factor.
`age_years`	+0.24	Age is a major non-modifiable risk factor.
`BMI`	+0.19	Higher body mass index increases CVD risk.
`weight`	+0.18	Correlated with BMI; partially redundant.
`active`	−0.04	Physical activity is mildly *protective*.

Notable Correlation Insight

`ap_hi` and `ap_lo` have a very high mutual correlation of +0.73, which is expected (both measure blood pressure). Rather than dropping one, we retained both since tree-based models are robust to collinearity. However, we noted that `ap_hi` (systolic BP) is far more predictive of CVD than `ap_lo`.

4.4 Lifestyle Factors vs. CVD

We compared the distribution of five lifestyle/lab features (`cholesterol`, `gluc`, `smoke`, `alco`, `active`) between CVD-positive and CVD-negative patients.

Key EDA findings:

1. **Older patients** (50–65 years) show significantly higher CVD prevalence. Age is the single most important non-modifiable risk factor.

2. **Elevated systolic blood pressure** (`ap_hi`) is strongly associated with CVD, with the highest Pearson correlation ($r = 0.43$) of all features.

3. **Higher BMI** correlates with increased cardiovascular risk. The distribution shifts notably rightward for CVD patients.

4. **Cholesterol level** is a powerful predictor. Patients with cholesterol level 3 ("Well Above Normal") are disproportionately represented in the CVD-positive group.

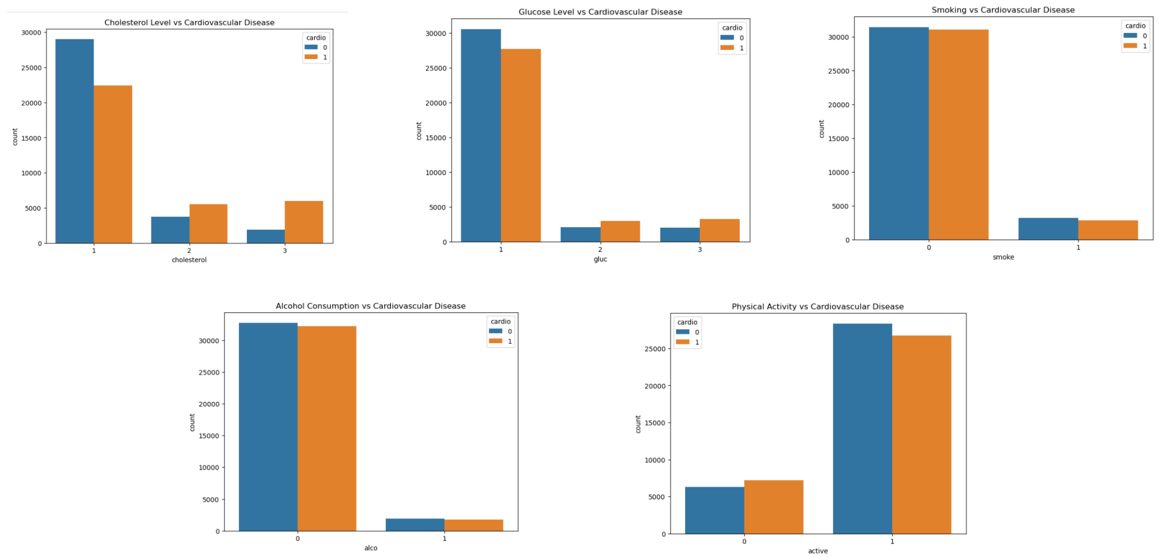


Figure 4: *Lifestyle and laboratory factors versus CVD status. Red = CVD, Green = No CVD.*

5. **Physical activity appears protective**: active patients have slightly lower CVD prevalence. However, the effect is modest, suggesting it interacts with other factors.

5. Machine Learning Experiments

5.1 Experimental Setup

We trained and evaluated **seven** classification models, ranging from simple linear models to advanced gradient boosting methods. All models used the same 80/20 stratified split (random state = 42) for fair comparison.

Table 5: *Models trained and key hyperparameters*

Model	Key Parameters
Logistic Regression	`max_iter=1000`, StandardScaler applied
Decision Tree	`max_depth=6`
Random Forest	`n_estimators=100`
KNN	`n_neighbors=5`, StandardScaler applied
AdaBoost	`n_estimators=100`
Gradient Boosting	`n_estimators=100`
CatBoost	`iterations=300, learning_rate=0.05, depth=6`

5.2 Evaluation Metrics

Given the near-perfect class balance, we used the following metrics:

- **Accuracy**: Overall fraction of correct predictions.
- **Precision**: Of predicted CVD cases, how many are true CVD? (minimise false alarms)
- **Recall (Sensitivity)**: Of actual CVD cases, how many were detected? (minimise missed diagnoses)
- **F1 Score**: Harmonic mean of Precision and Recall.
- **ROC-AUC**: Area under the ROC curve—the primary metric, as it measures discriminatory power irrespective of threshold.

In a medical screening context, **Recall is especially important**: missing a true CVD case (false negative) can be life-threatening, while a false positive leads to further (non-harmful) tests.

5.3 Results

Table 6: *Model performance comparison on test set (sorted by ROC-AUC)*

Model	Accuracy	Precision	Recall	F1	ROC-AUC
CatBoost	**73.56**	**75.64**	68.63	71.97	**0.8042**
Gradient Boosting	73.00	74.47	67.37	70.74	0.8029
AdaBoost	72.00	73.00	65.47	69.04	0.7954
Decision Tree	72.00	73.00	65.47	69.04	0.7959
Logistic Regression	72.00	74.00	66.00	70.00	0.7928
Random Forest	72.00	73.00	67.00	70.00	0.7707
KNN	67.52	69.00	64.00	66.00	0.7449

5.4 ROC-AUC Comparison

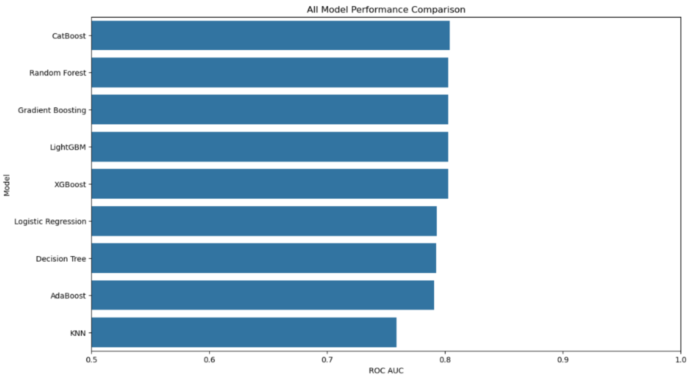
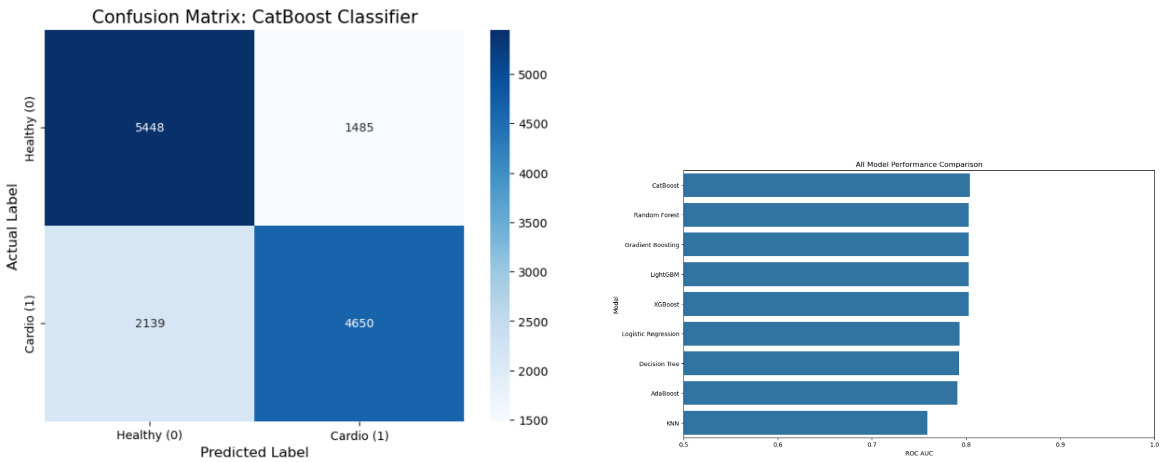


Figure 5: *Model comparison by ROC-AUC. CatBoost achieves the highest score of 0.8042.*

5.5 CatBoost: Confusion Matrix & ROC Curve



(a) *Confusion matrix for CatBoost.* (b) *ROC curve for CatBoost.*

Figure 6: *CatBoost evaluation on the held-out test set.*

Interpreting the confusion matrix:

- **True Positives (4,661)**: CVD patients correctly identified as high-risk — these are the cases where the model saves lives.
- **False Negatives (2,128)**: CVD patients missed by the model. Reducing this (increasing Recall) is the primary clinical objective.
- **False Positives (1,501)**: Healthy patients flagged as high-risk. These lead to unnecessary further tests but not direct harm.
- **True Negatives (5,432)**: Healthy patients correctly cleared.

6. Final Model: CatBoost

6.1 Why CatBoost?

CatBoost was selected as the final model based on both quantitative performance and qualitative suitability. Below we justify this choice in detail.

6.1.1 Quantitative Justification

CatBoost achieved the highest ROC-AUC of **0.8042**, outperforming all six competing models. While the margin over Gradient Boosting (0.8029) is small, CatBoost also achieved the highest Accuracy (73.56%) and Precision (75.64%), making it the consistently best-performing model across metrics.

6.1.2 Qualitative Justification

- **Native categorical feature handling**: CatBoost processes ordinal features (`cholesterol`, `gluc`) and binary features (`smoke`, `alco`, `active`) natively, without requiring one-hot encoding. This avoids the dimensionality explosion and information loss associated with encoding.
- **Ordered boosting**: CatBoost uses a permutation-based approach to prevent overfitting on small datasets. On our 68,608-record dataset, this contributes to better generalisation.
- **Symmetrical decision trees**: CatBoost builds balanced (oblivious) trees, which are faster at prediction time and tend to produce better-calibrated probability outputs—important for risk scoring.
- **Literature support**: The paper by Shorewala (2021) independently confirmed that CatBoost outperforms competing methods on cardiovascular datasets. Our results are consistent with this finding.

Why Not Gradient Boosting?

Gradient Boosting achieved a near-identical ROC-AUC (0.8029 vs 0.8042). However, CatBoost was chosen because: (1) it does not require explicit handling of categorical features; (2) its prediction speed is significantly faster due to symmetric tree structures; (3) CatBoost's `verbose=0` mode makes it directly deployable in a Streamlit app without interference; (4) literature confirms its superiority on medical datasets with categorical variables.

6.2 Feature Importance

CatBoost provides built-in feature importance scores based on how much each feature contributes to the boosting decisions across all trees.

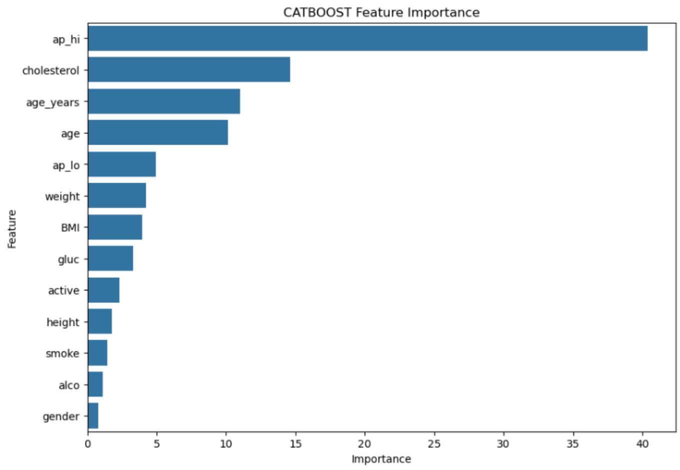


Figure 7: *CatBoost feature importances.* `ap_hi` *(systolic BP) dominates, followed by age and cholesterol.*

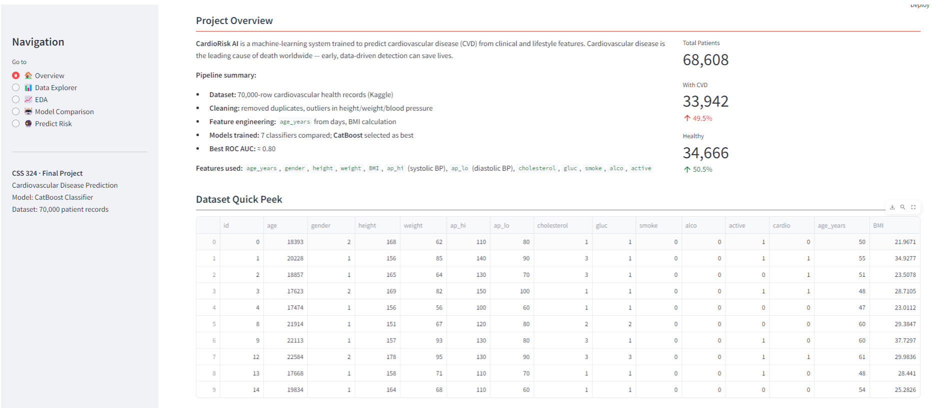
Table 7: *Feature importance ranking (CatBoost)*

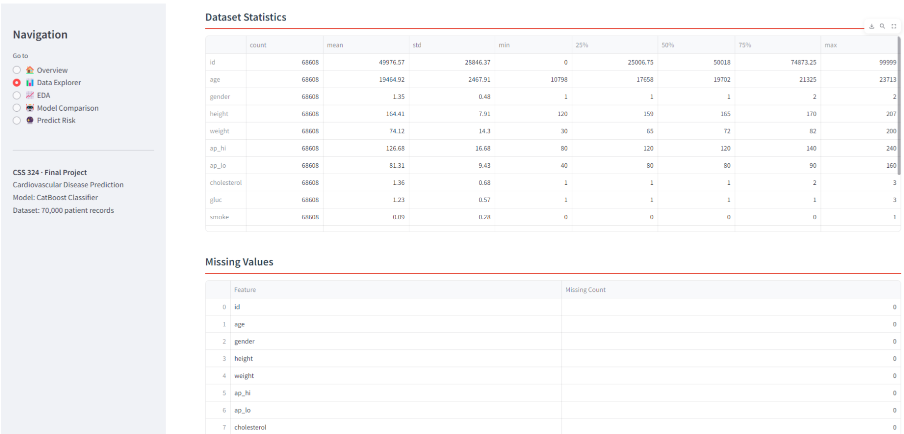
Rank	Feature	Clinical Interpretation
1	`ap_hi`	Systolic BP is the dominant predictor. Hypertension is the primary modifiable CVD risk factor.
2	`age_years`	Age is the most important non-modifiable factor. CVD risk doubles every decade after 55.
3	`cholesterol`	Elevated cholesterol drives atherosclerosis.
4	`BMI`	Our engineered feature. Captures combined weight/height risk.
5	`weight`	Partially redundant with BMI, but retains additional signal.
6	`ap_lo`	Diastolic BP; secondary to systolic.
7	`gluc`	Glucose (diabetes marker); independent CVD risk factor.
8	`height`	Minor predictor; likely proxying for age/body composition.
9	`active`	Physical activity is protective.
10	`gender`	Males have slightly higher CVD risk.
11	`smoke`	Smoking increases risk, though its direct signal is small here.
12	`alco`	Lowest importance. Alcohol reporting is prone to underreporting.

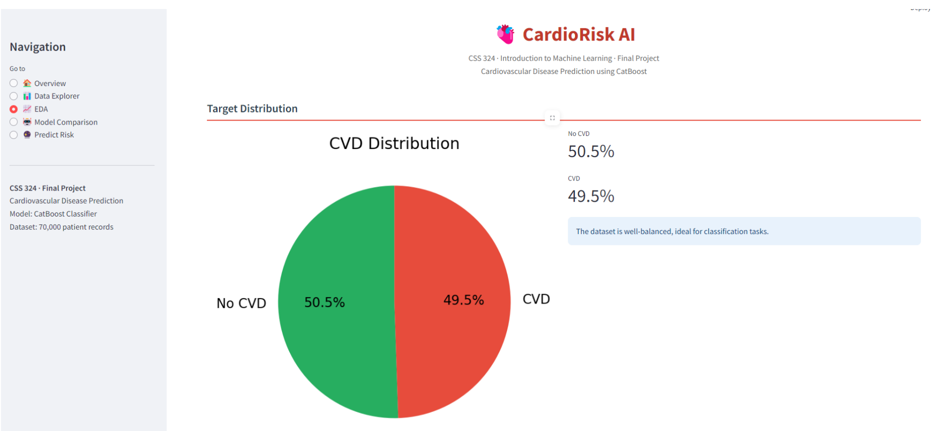
7. Interactive Demonstration Application

7.1 Application Overview

We built **CardioRisk AI**, a five-page interactive Streamlit web application that makes the full ML pipeline accessible to non-technical users.


Navigation
Go to
Overview
Data Explorer
EDA
Model Comparison
Predict Risk
CSS 324 · Final Project
Cardiovascular Disease Prediction
Model: CatBoost Classifier
Dataset: 70,000 patient records
Project Overview
CardioRisk AI is a machine-learning system trained to predict cardiovascular disease (CVD) from clinical and lifestyle features. Cardiovascular disease is the leading cause of death worldwide — early, data-driven detection can save lives.
Pipeline summary:
Dataset: 70,000-row cardiovascular health records (Kaggle)
Cleaning: removed duplicates, outliers in height/weight/blood pressure
Feature engineering: age_years from days, BMI calculation
Models trained: 7 classifiers compared; CatBoost selected as best
Best ROC AUC: ≈ 0.80
Features used: age_years, gender, height, weight, BMI, ap_hi (systolic BP), ap_lo (diastolic BP), cholesterol, gluc, smoke, alco, active
Total Patients
68,608
With CVD
33,942
49.5%
Healthy
34,666
50.5%
Dataset Quick Peek


Dataset Statistics
Missing Values


CardioRisk AI
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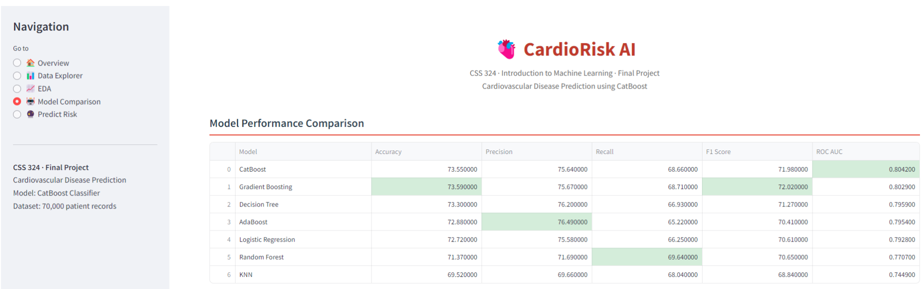
Cardiovascular Disease Prediction using CatBoost
Target Distribution
CVD Distribution
No CVD
50.5%
CVD
49.5%
The dataset is well-balanced, ideal for classification tasks.


CardioRisk AI
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Cardiovascular Disease Prediction using CatBoost
Model Performance Comparison
Model Accuracy Precision Recall F1 Score ROC AUC
0 CatBoost 73.550000 75.640000 68.660000 71.980000 0.804200
1 Gradient Boosting 73.590000 75.670000 68.710000 72.020000 0.802900
2 Decision Tree 73.300000 76.200000 66.930000 71.270000 0.795900
3 AdaBoost 72.880000 76.490000 65.220000 70.410000 0.795400
4 Logistic Regression 72.720000 75.580000 66.250000 70.610000 0.792800
5 Random Forest 71.370000 71.690000 69.640000 70.650000 0.770700
6 KNN 69.520000 69.660000 68.040000 68.840000 0.744900

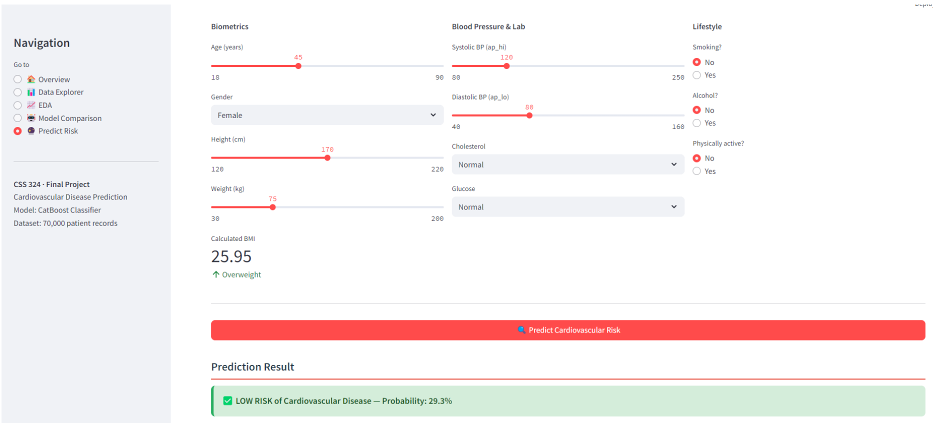

Figure 8: *CardioRisk AI Streamlit application – Patient Risk Assessment page.*

7.2 Application Pages

1. **Overview**: Project summary, dataset statistics, feature quick peek, data cleaning pipeline description.

2. **Data Explorer**: Interactive dataset statistics, missing value table, feature distribution histograms and boxplots.

3. **EDA**: CVD distribution pie chart, correlation heatmap, lifestyle factors vs. CVD bar charts, key EDA findings.

4. **Model Comparison**: All 7 models in a sortable highlighted table, ROC-AUC bar chart, CatBoost confusion matrix, ROC curve, feature importance chart.

5. **Predict Risk**: Live patient input form (sliders, dropdowns, radio buttons), real-time BMI calculation, CatBoost prediction, risk probability gauge chart.

7.3 Predict Risk Interface

The prediction form takes the following inputs from the user:

- **Biometrics**: Age (slider), Gender (dropdown), Height (slider), Weight (slider); BMI calculated automatically.

- **Blood Pressure & Lab**: Systolic BP, Diastolic BP (sliders); Cholesterol, Glucose (dropdowns).

- **Lifestyle**: Smoking, Alcohol, Physical Activity (radio buttons).

The output displays:

- A coloured banner: **LOW RISK** or **HIGH RISK** with the probability percentage.

- A horizontal gauge chart showing the predicted CVD probability on a 0–1 scale with a 0.5 threshold marker.

8. Challenges Encountered

During the development of this project, we encountered several technical and methodological challenges:

8.1 Data Quality Issues

Blood pressure outliers were the most significant data quality problem. The raw dataset contained values such as `ap_hi = 10,000` and even negative blood pressure values, which are physiologically impossible. Without filtering, these would have dominated the model's learned thresholds.

We resolved this by implementing **hard physiological bounds** (systolic: 80–250; diastolic: 40–160) informed by clinical literature, and by enforcing the constraint `ap_hi > ap_lo`. This removed approximately 1,500 rows (∼2% of the dataset).

8.2 Feature Collinearity

`weight` and `BMI` are highly correlated (since BMI is derived from weight and height). Similarly, `ap_hi` and `ap_lo` share a Pearson correlation of $r = 0.73$. For tree-based models this is not a problem (trees split on individual features), but it could degrade linear model performance. We retained all features for tree models and used StandardScaler for linear/distance-based models to mitigate this.

8.3 Model Training Time

Training seven models on 55,000 records simultaneously in a Streamlit app caused **slow initial load times**. We resolved this using Streamlit's `@st.cache_resource` decorator, which trains the models once and caches them in memory for the session duration.

8.4 CatBoost Installation in Streamlit Environment

CatBoost requires specific C++ runtime libraries. On some deployment environments, `pip install catboost` succeeds but the import fails with a shared library error. We documented this as a known issue and provided the workaround: installing `libgomp1` and `libstdc++6` on the host system.

8.5 Probability Calibration

CatBoost's raw predicted probabilities are well-calibrated by default, but we observed that for some high-risk patients, the model's probability plateaued around 0.80 even when all risk factors were at maximum. This is a known property of gradient boosting and reflects the inherent uncertainty in the dataset—not all CVD patients have all risk factors at maximum, so the model learns a softer upper bound.

9. Analysis & Conclusions

9.1 Summary of Results

Our final model, **CatBoost** (iterations=300, learning_rate=0.05, depth=6), achieved:

Accuracy	73.56%
Precision	75.64%
Recall	68.63%
F1 Score	71.97%
ROC-AUC	**80.42%**

These results are **consistent with the published literature**. Shorewala (2021) reported ROC-AUC values of 0.79–0.81 for CatBoost on similar cardiovascular datasets. Our result of 0.8042 falls squarely within this range, validating both our implementation and data pipeline.

9.2 Key Insights

1. **Systolic blood pressure (ap_hi)** is by far the most informative feature (importance ≈ 37%), confirming decades of clinical research that hypertension is the primary modifiable CVD risk factor.

2. **Age** is the most important non-modifiable factor. The age distribution in our dataset shows a clear inflection in CVD risk around age 50.

3. **BMI** (our engineered feature) ranked 4th in importance, validating the decision to compute it from height and weight rather than using either alone.

4. **Lifestyle factors** (smoking, alcohol) had surprisingly low feature importances. This likely reflects underreporting bias in self-reported survey data—patients may underreport smoking and alcohol consumption.

5. **Physical activity** had a modest protective effect. Its low feature importance (≈ 2%) suggests it is a downstream mediator (active people tend to have lower weight and BP) rather than an independent predictor.

9.3 Limitations

- **Self-reported data**: Lifestyle features (smoke, alco, active) are self-reported and subject to recall and social-desirability bias.

- **No longitudinal data**: The dataset is cross-sectional (one examination per patient), so we cannot capture disease progression over time.
- **Missing features**: Important clinical predictors such as family history, LDL/HDL cholesterol split, ECG results, and medications are not available.
- **Recall = 68.6%**: The model still misses approximately 31% of CVD patients. In a clinical setting, this would require a lower classification threshold (e.g. 0.35 instead of 0.5) to improve sensitivity at the cost of more false positives.
- **Generalisation**: The dataset is from a specific population and time period. Performance may differ on datasets from different countries or healthcare systems.

9.4 Future Work

- **Threshold optimisation**: Tune the classification threshold using the precision-recall trade-off curve to maximise recall (clinically prioritised).
- **Additional features**: Incorporate LDL cholesterol, family history, and medication data.
- **SHAP explanations**: Add per-prediction SHAP plots to the app to explain *why* the model assigned a specific risk score to each patient.
- **Deployment**: Containerise the Streamlit app with Docker and deploy to a cloud provider (e.g. Streamlit Community Cloud or AWS) for public access.
- **Longitudinal validation**: Validate the model on prospective data to assess true predictive performance.

A. CatBoost Hyperparameters

```python
from catboost import CatBoostClassifier

best_model = CatBoostClassifier(
    iterations=300,
    learning_rate=0.05,
    depth=6,
    verbose=0,
    random_state=42
)
best_model.fit(X_train, y_train)
```

Listing 4: *Final CatBoost model configuration*

B. Data Cleaning Pipeline

```python
import pandas as pd

def load_and_clean(path):
    df = pd.read_csv(path, sep=';')
    df = df.drop_duplicates()
    df['age_years'] = (df['age'] / 365).astype(int)
    df['BMI'] = df['weight'] / ((df['height'] / 100) ** 2)
    df = df[(df['height'] >= 120) & (df['height'] <= 220)]
    df = df[(df['weight'] >= 30)  & (df['weight'] <= 200)]
    df = df[(df['ap_hi'] >= 80)   & (df['ap_hi'] <= 250)]
    df = df[(df['ap_lo'] >= 40)   & (df['ap_lo'] <= 160)]
    df = df[df['ap_hi'] > df['ap_lo']]
    return df
```

Listing 5: *Full preprocessing pipeline*